

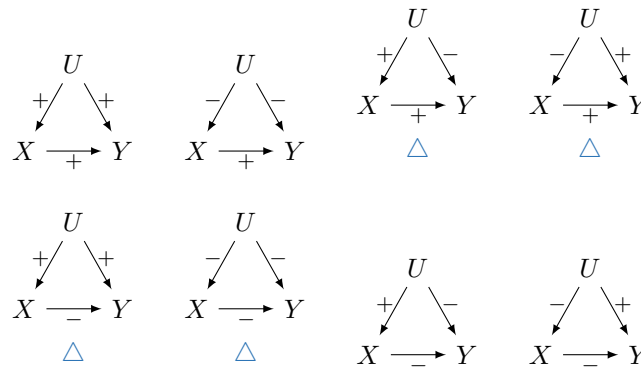
Quick Reference: Statistical Thinking

Humble

Causal Inference & Missing Data

1 Simpson's Paradox

After controlling for a third variable, the correlation between two variables can *disappear* or even *reverse*. With U a common cause of X and Y and a direct effect $X \rightarrow Y$, the eight sign patterns below show when the marginal and conditional associations can disagree.



Cases marked with “ Δ ” may lead to Simpson's Paradox.

Intuitively, the confounding bias travels along $X \leftarrow U \rightarrow Y$. When this bias *opposes* the sign of the direct effect $X \rightarrow Y$ (top row: U -arrows of *opposite* sign against a positive direct effect; bottom row: U -arrows of the *same* sign against a negative direct effect), the adjusted and unadjusted associations can carry different signs.

2 Potential Outcome Framework

Let $X \in \{0, 1\}$ be the treatment (1 = treated, 0 = control) and Y the outcome.

Assumption 1 [No Interference]. $Y_i(x_1, \dots, x_i, \dots, x_n) = Y_i(x_i)$.

Assumption 2 [No Multiple Versions]. There are no other versions of treatment.

Assumption 3 [Consistency]. The observed Y_i is the realisation of the potential outcome under the received treatment:

$$Y_i = X_i Y_i(1) + (1 - X_i) Y_i(0).$$

Together, Assumptions 1–3 constitute **SUTVA**.

Three causal estimands:

$$\underbrace{\tau_i = Y_i(1) - Y_i(0)}_{\text{Individual}} \quad \underbrace{\tau = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))}_{\text{Average (ACE)}} \quad \underbrace{\tau_u = \frac{\sum_{i=1}^n \mathbb{I}(U_i = u) \tau_i}{\sum_{i=1}^n \mathbb{I}(U_i = u)}}_{\text{Subgroup}}$$

3 Randomization

Random assignment makes treatment independent of the potential outcomes, $X \perp\!\!\!\perp \{Y(0), Y(1)\}$, which identifies the ACE from observed means:

$$\begin{aligned}\mathbb{E}(Y | X = 1) &= \mathbb{E}(XY(1) + (1 - X)Y(0) | X = 1) = \mathbb{E}(Y(1) | X = 1) = \mathbb{E}(Y(1)), \\ \mathbb{E}(Y | X = 0) &= \mathbb{E}(Y(0)) \quad (\text{similarly}),\end{aligned}$$

$$\tau = \mathbb{E}(Y(1)) - \mathbb{E}(Y(0)) = \mathbb{E}(Y | X = 1) - \mathbb{E}(Y | X = 0).$$

Exercise 1 — Randomization & identification of the ATE

Consider a completely randomized experiment with binary treatment X , where exactly n_1 of n units are assigned to treatment and $n_0 = n - n_1$ to control. Let the estimand be the finite-population average treatment effect

$$\tau_n = \frac{1}{n} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\},$$

and the difference-in-means estimator be $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$, where

$$\bar{Y}_1 = \frac{1}{n_1} \sum_{i=1}^n X_i Y_i, \quad \bar{Y}_0 = \frac{1}{n_0} \sum_{i=1}^n (1 - X_i) Y_i.$$

1. Derive $\hat{\tau} - \tau_n = (\bar{Y}_1(1) - \bar{Y}(1)) - (\bar{Y}_0(0) - \bar{Y}(0))$, where $\bar{Y}(x) = n^{-1} \sum_{i=1}^n Y_i(x)$.
2. Derive the finite-population variance $\text{Var}(\hat{\tau} | \{Y_i(1), Y_i(0)\}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{n}$, with S_1^2, S_0^2, S_τ^2 the finite-population variances of $Y_i(1)$, $Y_i(0)$ and $Y_i(1) - Y_i(0)$.
3. Explain why S_τ^2 is not identified from the observed data.
4. Show that the Neyman variance estimator $\hat{V}_N = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}$ is conservative.

Solution

1.

$$\hat{\tau} - \tau_n = (\bar{Y}_1 - \bar{Y}_0) - \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)) = \frac{1}{n_1} \sum_{i=1}^n X_i Y_i - \frac{1}{n_0} \sum_{i=1}^n (1 - X_i) Y_i - \bar{Y}(1) + \bar{Y}(0).$$

By consistency $X_i Y_i = X_i Y_i(1)$ and $(1 - X_i) Y_i = (1 - X_i) Y_i(0)$. Defining $\bar{Y}_1(1) = \frac{1}{n_1} \sum_i X_i Y_i(1)$ and $\bar{Y}_0(0) = \frac{1}{n_0} \sum_i (1 - X_i) Y_i(0)$,

$$\hat{\tau} - \tau_n = (\bar{Y}_1(1) - \bar{Y}(1)) - (\bar{Y}_0(0) - \bar{Y}(0)).$$

2. From part 1, $\text{Var}(\hat{\tau}) = \mathbb{E}(\hat{\tau} - \tau_n)^2$ expands as

$$\mathbb{E}(\bar{Y}_1(1) - \bar{Y}(1))^2 + \mathbb{E}(\bar{Y}_0(0) - \bar{Y}(0))^2 - 2\mathbb{E}(\bar{Y}_1(1) - \bar{Y}(1))(\bar{Y}_0(0) - \bar{Y}(0)).$$

The assignment moments are

$$\mathbb{E}X_i = \mathbb{E}X_i^2 = \frac{n_1}{n}, \quad \text{Var } X_i = \frac{n_0 n_1}{n^2}, \quad \mathbb{E}X_i X_j = \frac{n_1(n_1 - 1)}{n(n - 1)}, \quad \text{Cov}(X_i, X_j) = -\frac{n_0 n_1}{n^2(n - 1)} \quad (i \neq j).$$

First term.

$$\begin{aligned}\mathbb{E}(\bar{Y}_1(1) - \bar{Y}(1))^2 &= \text{Var}\left(\frac{1}{n_1} \sum_i X_i Y_i(1)\right) = \frac{1}{n_1^2} \left[\sum_i Y_i^2(1) \text{Var } X_i + 2 \sum_{i < j} Y_i(1) Y_j(1) \text{Cov}(X_i, X_j) \right] \\ &= \frac{n_0}{n(n-1)n_1} \sum_{i=1}^n (Y_i(1) - \bar{Y}(1))^2 = \frac{n_0}{n n_1} S_1^2 = \left(\frac{1}{n_1} - \frac{1}{n}\right) S_1^2.\end{aligned}$$

Second term. Symmetrically,

$$\mathbb{E}(\bar{Y}_0(0) - \bar{Y}(0))^2 = \text{Var}\left(\frac{1}{n_0} \sum_i (1 - X_i) Y_i(0)\right) = \frac{n_1}{n n_0} S_0^2 = \left(\frac{1}{n_0} - \frac{1}{n}\right) S_0^2.$$

Cross term.

$$\begin{aligned}\mathbb{E}(\bar{Y}_1(1) - \bar{Y}(1))(\bar{Y}_0(0) - \bar{Y}(0)) &= \text{Cov}\left(\frac{1}{n_1} \sum_i X_i Y_i(1), \frac{1}{n_0} \sum_i (1 - X_i) Y_i(0)\right) \\ &= \frac{-1}{n_0 n_1} \left[\sum_i Y_i(0) Y_i(1) \text{Var } X_i + 2 \sum_{i < j} Y_i(0) Y_j(1) \text{Cov}(X_i, X_j) \right] = -\frac{1}{n} S_{10},\end{aligned}$$

where $S_{10} = \frac{1}{n-1} \sum_i (Y_i(1) - \bar{Y}(1))(Y_i(0) - \bar{Y}(0))$. Since $S_\tau^2 = S_1^2 - 2S_{10} + S_0^2$, combining the three pieces gives

$$\text{Var}(\hat{\tau} \mid \{Y_i(0), Y_i(1)\}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{n}.$$

3. $Y_i(1)$ and $Y_i(0)$ are never observed simultaneously, so the joint distribution—and hence S_τ^2 , which depends on it—cannot be identified from the data.

4. Because $\mathbb{E}\hat{V}_N - \text{Var } \hat{\tau} = \frac{S_\tau^2}{n} \geq 0$, the Neyman estimator overstates the variance on average and is therefore conservative.

4 Observational Study

We retain the potential-outcome framework. Here Z denotes the treatment indicator and X the observed covariate.

Assumption 4 [Ignorability]. $(Y(1), Y(0)) \perp\!\!\!\perp Z \mid X$.

Definition (Propensity Score). The propensity score is $e(X, Y(1), Y(0)) = \mathbb{P}(Z = 1 \mid X, Y(1), Y(0))$. Under ignorability it collapses to

$$e(X) = \mathbb{P}(Z = 1 \mid X).$$

Theorem. Under ignorability, $(Y(1), Y(0)) \perp\!\!\!\perp Z \mid e(X)$; the propensity score is a balancing score and can be used for stratification:

$$\text{ACE} = \int \mathbb{E}(Y(1) - Y(0) \mid e(X)) p_{e(X)}(x) dx \approx \sum_{k=1}^K \mathbb{P}(e(X) \in I_k) \mathbb{E}(Y(1) - Y(0) \mid e(X) \in I_k).$$

5 IPW — Inverse Probability Weighting

Exercise 2 — IPW identification

Assume consistency, ignorability and positivity. Let $e(V) = \mathbb{P}(X = 1 \mid V)$ and define

$$\hat{\mu}_1^{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \frac{X_i Y_i}{\hat{e}(V_i)}, \quad \hat{\mu}_0^{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \frac{(1 - X_i) Y_i}{1 - \hat{e}(V_i)}, \quad \hat{\tau}^{\text{IPW}} = \hat{\mu}_1^{\text{IPW}} - \hat{\mu}_0^{\text{IPW}}.$$

1. Show that if the true $e(V)$ is known then $\mathbb{E}\left\{\frac{XY}{e(V)}\right\} = \mathbb{E}\{Y(1)\}$ and $\mathbb{E}\left\{\frac{(1-X)Y}{1-e(V)}\right\} = \mathbb{E}\{Y(0)\}$.
2. Derive the same result via iterated expectation and consistency.
3. Explain why IPW becomes unstable when positivity is nearly violated.
4. Compare the Horvitz–Thompson form $\frac{1}{n} \sum_i \frac{X_i Y_i}{\hat{e}(V_i)}$ with the Hájek form $\frac{\sum_i X_i Y_i / \hat{e}(V_i)}{\sum_i X_i / \hat{e}(V_i)}$.

Solution

(1& 2) By consistency $Y = XY(1) + (1-X)Y(0)$, so $XY = XY(1)$. With $e(V) = \mathbb{E}(X|V)$ and ignorability $X \perp\!\!\!\perp Y(1)|V$,

$$\begin{aligned} \mathbb{E}\left(\frac{XY}{e(V)}\right) &= \mathbb{E}\left(\frac{XY(1)}{\mathbb{E}(X|V)}\right) = \mathbb{E}\left[\mathbb{E}\left(\frac{XY(1)}{\mathbb{E}(X|V)} \mid V\right)\right] \\ &= \mathbb{E}\left[\mathbb{E}(X|V) \mathbb{E}(Y(1)|V) \cdot \frac{1}{\mathbb{E}(X|V)}\right] = \mathbb{E}[\mathbb{E}(Y(1)|V)] = \mathbb{E}\{Y(1)\}. \end{aligned}$$

Similarly $\mathbb{E}\left\{\frac{(1-X)Y}{1-e(V)}\right\} = \mathbb{E}\{Y(0)\}$.

(3) If $e(V) \rightarrow 0$ or 1 for some units, the weights $1/\hat{e}(V)$ blow up; a few extreme weights dominate the sum, inflating the variance and making the estimator unstable.

(4) The Hájek form normalises the weights to sum to one. It is slightly biased in finite samples but is location-invariant and far more stable than the Horvitz–Thompson form, which is unbiased but high-variance when weights are extreme.

6 Outcome Regression

Exercise 3 — g-formula

Suppose treatment is not randomized. Assume $(Y(1), Y(0)) \perp\!\!\!\perp X | V$ and positivity $0 < \mathbb{P}(X = 1 | V) < 1$ a.s. Target $\tau = \mathbb{E}\{Y(1) - Y(0)\}$.

1. Prove $\mathbb{E}\{Y(x)\} = \mathbb{E}[\mathbb{E}(Y | X = x, V)]$, $x = 0, 1$.
2. Deduce the g-formula $\tau = \mathbb{E}[m_1(V) - m_0(V)]$.
3. Explain why positivity is needed for the g-formula to be meaningful.

Solution

1. By consistency and then ignorability,

$$\mathbb{E}[\mathbb{E}(Y | X = x, V)] = \mathbb{E}[\mathbb{E}(Y(x) | X = x, V)] = \mathbb{E}[\mathbb{E}(Y(x) | V)] = \mathbb{E}\{Y(x)\}.$$

2. With $m_x(V) := \mathbb{E}(Y | X = x, V)$ the g-formula follows directly from part 1: $\tau = \mathbb{E}\{Y(1)\} - \mathbb{E}\{Y(0)\} = \mathbb{E}[m_1(V) - m_0(V)]$.
3. If $\mathbb{P}(X = x | V = v) = 0$ on a region, $m_x(v)$ is not defined there (no data), so $\mathbb{E}[m_x(V)]$ cannot be evaluated; positivity guarantees both treatment arms are observed at every covariate value.

OR procedure. (i) fit a model for $\mathbb{E}(Y | X, V) = m_x(V)$; (ii) estimate $\widehat{\text{ACE}}_{\text{OR}} = \frac{1}{n} \sum_{i=1}^n \{\hat{m}_1(V_i) - \hat{m}_0(V_i)\}$.

Its correctness depends *strongly* on correct specification of the nuisance model m_x .

7 Double Robust Estimation (AIPW)

Exercise 4 — Double robustness of AIPW

Assume consistency, ignorability and positivity; target $\tau = \mathbb{E}\{Y(1) - Y(0)\}$. Estimate regressions $m_1(V), m_0(V)$ and propensity score $e(V)$, and define

$$\hat{\tau}^{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{m}_1(V_i) - \hat{m}_0(V_i) + \frac{X_i \{Y_i - \hat{m}_1(V_i)\}}{\hat{e}(V_i)} - \frac{(1 - X_i) \{Y_i - \hat{m}_0(V_i)\}}{1 - \hat{e}(V_i)} \right].$$

1. Show that if $e(V)$ is correct then the population analogue equals τ for *any* fixed $\hat{m}_1, \hat{m}_0$.
2. Show that if m_1, m_0 are correct then it equals τ for *any* fixed $\hat{e}$ with $0 < \hat{e} < 1$.
3. Conclude that the AIPW estimating function is doubly robust.

Solution

1. (**propensity score correct**, $e(V) = \mathbb{P}(X = 1 | V)$.)

$$\begin{aligned} \mathbb{E} \left(\frac{X \{Y - \hat{m}_1(V)\}}{e(V)} \right) &= \mathbb{E} \left[\mathbb{E} \left(\frac{X \{Y(1) - \hat{m}_1(V)\}}{e(V)} \mid V \right) \right] = \mathbb{E} \left[\frac{Y(1) - \hat{m}_1(V)}{e(V)} \mathbb{E}(X | V) \right] \\ &= \mathbb{E}[Y(1) - \hat{m}_1(V)] = \mathbb{E}\{Y(1)\} - \mathbb{E}\hat{m}_1(V). \end{aligned}$$

Symmetrically $\mathbb{E} \left(\frac{(1-X) \{Y - \hat{m}_0(V)\}}{1-e(V)} \right) = \mathbb{E}\{Y(0)\} - \mathbb{E}\hat{m}_0(V)$. Adding $\mathbb{E}[\hat{m}_1(V) - \hat{m}_0(V)]$ the $\hat{m}$ -terms cancel, leaving $\mathbb{E}\{Y(1)\} - \mathbb{E}\{Y(0)\} = \tau$.

2. (**outcome regression correct**, $m_x(V) = \mathbb{E}(Y | X = x, V) = \mathbb{E}(Y(x) | V)$.) The augmentation term has zero mean:

$$\mathbb{E} \left(\frac{X \{Y - m_1(V)\}}{\hat{e}(V)} \right) = \mathbb{E} \left(\frac{X \{Y(1) - \mathbb{E}(Y(1) | V)\}}{\hat{e}(V)} \right),$$

and conditioning on V , ignorability gives $\mathbb{E}(X \{Y(1) - \mathbb{E}(Y(1) | V)\} | V) = \mathbb{E}(X | V) \mathbb{E}(Y(1) | V) - \mathbb{E}(X | V) \mathbb{E}(Y(1) | V) = 0$, forcing the whole term to 0. Hence only $\mathbb{E}[m_1(V) - m_0(V)] = \tau$ survives.

3. Each part needs only *one* nuisance model correct, so the estimating function is unbiased as long as the propensity score *or* the outcome regression is correctly specified — this is double robustness.

Two equivalent readings of each $\hat{\mu}$:

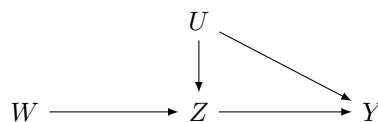
$$\hat{\mu}_1 = \underbrace{\mathbb{E} \left(\frac{XY}{e(V)} \right)}_{\text{IPW}} + \underbrace{\mathbb{E} \left[\left(1 - \frac{X}{e(V)} \right) \hat{m}_1(V) \right]}_{\text{augmentation}} \quad (\text{Augmented IPW})$$

$$\hat{\mu}_1 = \underbrace{\mathbb{E}(m_1(V))}_{\text{OR}} + \underbrace{\mathbb{E} \left(\frac{X \{Y - m_1(V)\}}{e(V)} \right)}_{\text{residual correction}} \quad (\text{Bias-corrected OR})$$

$\hat{\mu}_0$ is interpreted symmetrically.

8 Instrumental Variable

Basic case



A valid instrument W must satisfy

$$\textcircled{1} \quad W \perp\!\!\!\perp Y \mid (Z, U), \quad \textcircled{2} \quad W \not\perp\!\!\!\perp Y, \quad \textcircled{3} \quad W \perp\!\!\!\perp U.$$

In words: W affects the outcome Y *through and only through* Z . Writing $Y = \beta_0 + \beta_1 Z + U$, the moment condition $\text{Cov}(W, U) = 0$ yields

$$\beta_1 = \frac{\text{Cov}(W, Y)}{\text{Cov}(W, Z)}.$$

2SLS — Two-stage least squares

$$\text{Model } Y_i = \underbrace{X_i \beta}_{\text{explanatory}} + \underbrace{W_i^\top \nu}_{\text{exogenous controls}} + u_i.$$

- **Stage 1:** regress X_i on Z_i and $W_i \Rightarrow$ fitted value $\hat{X}_i$.
- **Stage 2:** regress Y_i on $\hat{X}_i$ and W_i .

Exercise 5 — Wald estimand / LATE

Let $Z \in \{0, 1\}$ be a binary instrument for a binary treatment $X \in \{0, 1\}$, with potential treatments $X(1), X(0)$ and potential outcomes $Y(x)$. Assume **(1) Relevance** $\mathbb{P}\{X(1) \neq X(0)\} > 0$; **(2) Independence** $Z \perp\!\!\!\perp \{Y(1), Y(0), X(1), X(0)\}$; **(3) Exclusion** $Y(z, x) = Y(x)$; **(4) Monotonicity** $X(1) \geq X(0)$ a.s.

1. Define the four principal strata.
2. Prove the Wald identity $\frac{\mathbb{E}(Y \mid Z = 1) - \mathbb{E}(Y \mid Z = 0)}{\mathbb{E}(X \mid Z = 1) - \mathbb{E}(X \mid Z = 0)} = \mathbb{E}\{Y(1) - Y(0) \mid X(1) > X(0)\}$ and interpret it.

Solution

1.

$$\begin{array}{ll} \text{Always-taker:} & X(1) = 1, X(0) = 1 \\ \text{Complier:} & X(1) = 1, X(0) = 0 \end{array} \quad \begin{array}{ll} \text{Never-taker:} & X(1) = 0, X(0) = 0 \\ \text{Defier:} & X(1) = 0, X(0) = 1 \end{array}$$

Monotonicity excludes the existence of defiers.

2. By exclusion and consistency, $Y(z) = Y(X(z)) = Y(1)X(z) + Y(0)(1 - X(z))$.

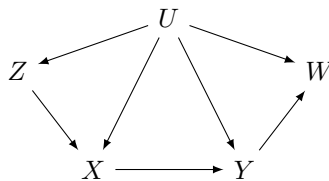
Numerator. By independence, $\mathbb{E}(Y \mid Z = 1) = \mathbb{E}(Y(X(1)))$ and $\mathbb{E}(Y \mid Z = 0) = \mathbb{E}(Y(X(0)))$, so

$$\begin{aligned} \mathbb{E}(Y \mid Z = 1) - \mathbb{E}(Y \mid Z = 0) &= \mathbb{E}(Y(X(1)) - Y(X(0))) = \mathbb{E}\{[Y(1) - Y(0)][X(1) - X(0)]\} \\ &= \mathbb{P}(X(1) > X(0)) \mathbb{E}\{Y(1) - Y(0) \mid X(1) > X(0)\}. \end{aligned}$$

Denominator. Likewise $\mathbb{E}(X \mid Z = 1) - \mathbb{E}(X \mid Z = 0) = \mathbb{E}(X(1) - X(0))$. By monotonicity $X(1) - X(0) \in \{0, 1\}$, hence it equals $\mathbb{P}(X(1) > X(0))$.

Dividing, the ratio equals $\mathbb{E}\{Y(1) - Y(0) \mid X(1) > X(0)\}$: the **Local Average Treatment Effect**, i.e. the ATE *among compliers*.

9 Double Negative Control



Exercise 6 — Proximal identification

Treatment X may be confounded by unobserved U ; Z is a negative-control exposure, W a negative-control outcome, V observed covariates (all discrete). The causal intuition is that U may affect X, Y, Z, W , but

$$Z \perp\!\!\!\perp (W, Y) \mid (U, X, V), \quad X \perp\!\!\!\perp W \mid (U, V).$$

1. Explain the meaning of the negative-control assumptions.
2. Under latent ignorability $Y(x) \perp\!\!\!\perp X \mid (U, V)$, derive the observable integral equation $p(y \mid x, z, v) = \sum_w h(y, x, v, w) p(w \mid x, z, v)$.
3. Show that a unique solution h identifies $p\{Y(x) = y \mid V = v\}$ by integrating h against $p(w \mid v)$.

Solution

1. Z is the negative-control *exposure*: it is driven by U , yet once we condition on U (and X) it must be unrelated to W and Y . W is the negative-control *outcome*: similarly driven by U , yet once we condition on U it must be unrelated to X . They act as proxies that “probe” the hidden confounder.
2. Introduce an *outcome confounding bridge function* h with

$$p(Y(x) \mid U, V) = \mathbb{E}(h(y, x, v, w) \mid U, V) = \sum_w h(y, x, v, w) p(w \mid U, V).$$

Then

$$\begin{aligned} p(y \mid x, z, v) &= \sum_u p(y \mid x, z, u, v) p(u \mid x, z, v) = \sum_u p(Y(x) \mid U, V) p(u \mid x, z, v) \\ &= \sum_{u, w} h(y, x, v, w) p(w \mid u, v) p(u \mid x, z, v). \end{aligned}$$

Under the negative-control assumptions $p(w \mid U, V) = p(w \mid x, u, v) = p(w \mid x, z, u, v)$, so the inner sum collapses: $\sum_u p(w \mid x, z, u, v) p(u \mid x, z, v) = p(w \mid x, z, v)$. Therefore

$$p(y \mid x, z, v) = \sum_w h(y, x, v, w) p(w \mid x, z, v).$$

This is an observable equation in which only h is unknown: the bridge function recovers the effect of U from the proxies Z, W .

3. Given such an h ,

$$p\{Y(x) = y \mid v\} = \sum_u p(Y(x) = y \mid u, v) p(u \mid v) = \sum_{u, w} h(y, x, v, w) p(w \mid u, v) p(u \mid v) = \sum_w h(y, x, v, w) p(w \mid v).$$

10 Missing Data

Notation. Y = outcome (possibly missing), X = fully observed covariates, R = missingness indicator with $R = 1$ observed, $R = 0$ unobserved.

Missing at Random (MAR)

The benchmark assumption $R \perp\!\!\!\perp Y \mid X$ (analogous to ignorability), under which causal-inference machinery transfers directly. Four standard approaches:

- ① Imputation $f(Y \mid X, R=0) = f(Y \mid X, R=1)$; ② IPW; ③ Outcome Regression; ④ Doubly Robust.

Missing Not at Random (MNAR)

① **A non-identifiability counterexample.** Let $Y \sim N(\mu, \sigma^2)$ and $\mathbb{P}(R = 1 | Y = y) = \frac{1}{1 + e^{-(\alpha + \beta y)}}$. Only $\mathbb{P}(Y, R = 1)$ is observed. The two parameter sets

$$\{N(1, 1), \alpha = -\frac{3}{2}, \beta = 1\} \quad \text{vs.} \quad \{N(2, 1), \alpha = \frac{3}{2}, \beta = -1\}$$

yield the *same* observed law $\mathbb{P}(Y, R = 1)$. Hence $\mathbb{P}(R = 1, Y)$ is **non-identifiable**.

Exercise 7 — Heckman selection model

Consider $Y = X^\top \beta + \varepsilon$ and $R = \mathbb{I}\{X^\top \gamma + \alpha Y + U > 0\}$, where (ε, U) is bivariate normal with mean zero, $\text{Var}(\varepsilon) = \sigma^2$, $\text{Var}(U) = 1$, $\text{Corr}(\varepsilon, U) = \rho$.

1. Explain why the mechanism is generally not MAR when $\alpha \neq 0$.
2. Show $\mathbb{P}(R = 1 | X, Y) = \Phi(X^\top \tilde{\gamma} + \tilde{\alpha} Y)$ and derive $\tilde{\gamma}, \tilde{\alpha}$.
3. Derive $Y | X, R = 1$ and explain how to estimate β .

Solution

1. When $\alpha \neq 0$, for fixed X the value of R depends on Y , so $Y \perp\!\!\!\perp R | X$ fails: the data are MNAR.
2. Write $\mathbb{P}(R = 1 | X, Y) = \mathbb{P}(U > -X^\top \gamma - \alpha Y | \varepsilon = Y - X^\top \beta)$. Since $\begin{pmatrix} U \\ \varepsilon \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho\sigma \\ \rho\sigma & \sigma^2 \end{pmatrix}\right)$, we have $U | \varepsilon \sim N\left(\frac{\rho}{\sigma}\varepsilon, 1 - \rho^2\right)$, hence

$$\mathbb{P}(R = 1 | X, Y) = 1 - \Phi\left(\frac{-X^\top \gamma - \alpha Y - \frac{\rho}{\sigma}\varepsilon}{\sqrt{1 - \rho^2}}\right) = \Phi\left(\frac{X^\top \gamma + \alpha Y + \frac{\rho}{\sigma}(Y - X^\top \beta)}{\sqrt{1 - \rho^2}}\right).$$

Matching to $\Phi(X^\top \tilde{\gamma} + \tilde{\alpha} Y)$ gives

$$\tilde{\gamma} = \frac{\gamma - \frac{\rho}{\sigma}\beta}{\sqrt{1 - \rho^2}}, \quad \tilde{\alpha} = \frac{\alpha + \frac{\rho}{\sigma}}{\sqrt{1 - \rho^2}}.$$

3. By Bayes' rule, $p(y | X, R = 1) = \frac{p(y | X) \mathbb{P}(R = 1 | X, Y)}{\mathbb{P}(R = 1 | X)}$, where $Y | X \sim N(X^\top \beta, \sigma^2)$ so $p(y | X) = \frac{1}{\sigma} \phi\left(\frac{y - X^\top \beta}{\sigma}\right)$. For the denominator,

$$\mathbb{P}(R = 1 | X) = \mathbb{P}(X^\top \gamma + \alpha X^\top \beta + \alpha \varepsilon + U > 0 | X), \quad \alpha \varepsilon + U \sim N(0, \alpha^2 \sigma^2 + 2\alpha \rho \sigma + 1),$$

$$\mathbb{P}(R = 1 | X) = \Phi\left(\frac{X^\top \gamma + \alpha X^\top \beta}{\sqrt{\alpha^2 \sigma^2 + 2\alpha \rho \sigma + 1}}\right).$$

The selected density $p(y | X, R = 1)$ is then fully parametric in $(\beta, \sigma, \gamma, \alpha, \rho)$, so β can be estimated by maximum likelihood on the observed sample (equivalently, Heckman's two-step inverse-Mills-ratio correction).